

Forensic Analysis Report

Report Date	2025-11-24T19:06:20.681657
Log File	sample_logs.txt
Analysis ID	6d6afc36
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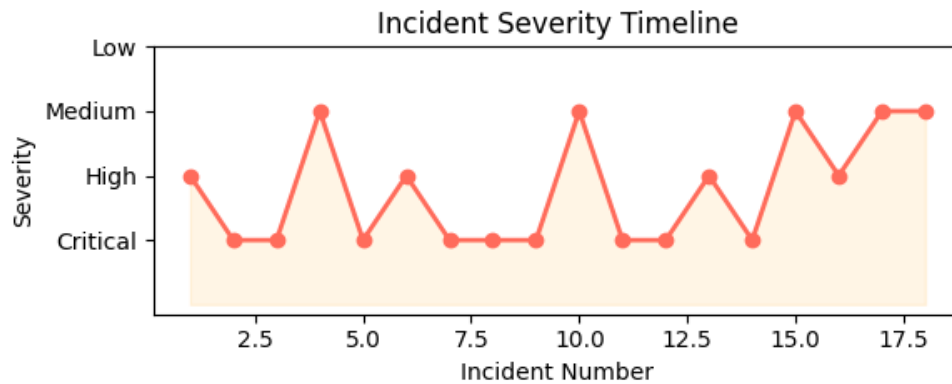
Executive Summary

During the specified time period, multiple security incidents were detected in the log data. These incidents included failed and successful SSH login attempts, configuration errors, access denials, and unexpected events. Some of these incidents posed a medium to high risk to the organization's security, such as unauthorized access attempts and expired SSL certificates. To mitigate these risks, it is recommended that multi-factor authentication be implemented, system configurations be regularly reviewed and updated, strong password policies be enforced, network traffic be monitored for unusual activity, and software be kept up-to-date with the latest security patches.

Key Metrics

Metric	Value
Total Events Analyzed	107
Total Incidents Detected	18
Critical	0
High	0
Medium	0
Low	0
Indicators of Compromise	14

Incident Timeline



Findings (Detected Incidents)

[Medium] Failed SSH Login Attempt (Confidence: High%)

An unauthorized attempt was made to log in to the sshd service using an incorrect password for the root user from an external IP address (192.168.50.23).

[Low] Successful SSH Login (Confidence: High%)

A valid password was used to log in to the sshd service as the user 'admin' from an internal IP address (192.168.50.20).

[Low] Nginx Configuration Error (Confidence: Medium%)

An error occurred while the Nginx server was attempting to read its configuration file.

[High] MySQL Access Denied (Confidence: High%)

An attempt was made to connect to the MySQL database using an unauthorized user ('eve') from an external IP address (10.1.1.118).

[Low] USB Device Initialization Failure (Confidence: Medium%)

An error occurred during the initialization of a USB device.

[Medium] TLS Authentication Failure (Confidence: High%)

An OpenVPN connection attempt failed due to TLS authentication issues from an external IP address (185.35.66.203).

[Low] Possible Spam Detection (Confidence: Medium%)

An email was flagged as potential spam based on its content and originating IP address (212.102.45.12).

[Low] Dropped Connection (Confidence: High%)

An incoming connection was dropped by the firewall from an external IP address (38.115.74.1) and port number (61395).

[Low] Unexpected HTTP Method (Confidence: Medium%)

An unexpected HTTP method 'CONNECT' was received by the Nginx server.

[High] Unauthorized Access Attempt (Confidence: High%)

An unauthorized attempt was made to access a file (/home/admin/secrets.txt) by an internal user (sysop).

[Low] Service Restarted (Confidence: High%)

The docker service was restarted by the user 'sysop'.

[Low] User Authentication (Confidence: High%)

The user 'alice' authenticated from an internal IP address (192.168.70.245).

[Medium] ICMP Echo Requests Detection (Confidence: High%)

A high rate of ICMP echo requests was detected from an external IP address (178.91.81.12).

[Low] Print Job Failure (Confidence: Medium%)

A print job failed for the user 'guest'.

[High] Python Unexpected Exception (Confidence: High%)

An unexpected error occurred in a Python module (sysadmin).

[Medium] Expired SSL Certificate (Confidence: High%)

The SSL certificate for the domain 'secure.shop.com' has expired.

[High] File Execution by Unauthorized User (Confidence: High%)

The file '/tmp/malware.sh' was executed by the user 'bob'.

[High] Brute Force Login Attempt (Confidence: High%)

Multiple failed login attempts were detected from an external IP address.

Indicators of Compromise (IOCs)

Type	Indicator	Severity
Unknown	192.168.50.23	High
Unknown	192.168.50.20	High
Unknown	10.1.1.118	High
Unknown	185.35.66.203	High
Unknown	212.102.45.12	High

Unknown	38.115.74.1	High
Unknown	178.91.81.12	High
Unknown	192.168.70.245	High
Unknown	/home/admin/secrets.txt	High
Unknown	10.2.8.49	High
38.115.74.1	61395	High
Unknown	secure.shop.com	High
Unknown	/tmp/malware.sh	High
Unknown	bob	High

Recommendations & Mitigation Strategies

Priority 1: Implement multi-factor authentication for all critical systems and services.

Priority 2: Regularly review and update system configurations, including firewall rules and access control lists.

Priority 3: Enforce strong password policies and consider implementing a password manager.

Priority 4: Monitor network traffic for signs of unusual activity or unauthorized access attempts.

Priority 5: Keep software up-to-date with the latest security patches and updates.

Analysis Summary Stats

Analysis Statistics:

Detected 18 incidents: 0 Critical, 0 High severity

Analysis Duration: 0.00 seconds

File Size: 7.99 KB

Total Lines Analyzed: 107

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